

Executive Summary

Customer Engagement & Product Utilization Analytics for Retention Strategy

Project Overview

Customer retention remains a critical challenge in the banking industry, where customer attrition directly impacts profitability, customer lifetime value, and long-term growth. This project analyzes the relationship between customer engagement, product utilization, and churn behavior within a European banking environment. The objective was to identify key behavioral and financial factors influencing customer retention and to provide actionable recommendations for improving customer loyalty.

Key Objectives

- Measure overall customer churn levels.
- Analyze the impact of customer engagement on retention.
- Evaluate how product utilization influences churn behavior.
- Identify customer segments with elevated churn risk.
- Support data-driven retention strategy development.

Methodology

The analysis was conducted using a customer-level banking dataset containing demographic, financial, and behavioral attributes. Data preparation included validation, cleaning, and exploratory analysis. Key performance indicators were calculated and visualized through an interactive Streamlit dashboard, enabling dynamic exploration of customer behavior patterns.

Major Findings

- The overall customer churn rate was approximately **20.37%**, indicating a significant retention challenge.
- Customers with lower engagement levels were substantially more likely to exit the bank.
- Customers utilizing a limited number of banking products exhibited higher churn rates compared to customers with broader product adoption.
- Active members demonstrated considerably stronger retention than inactive customers.
- Geography, age, account balance, and tenure were found to influence churn behavior across different customer segments.

- High-value customers represented a meaningful portion of churned accounts, highlighting potential revenue risks.

Recommendations

Based on the analysis, the following strategies are recommended:

- Increase customer engagement through personalized communication and loyalty initiatives.
- Promote cross-selling opportunities to encourage adoption of additional banking products.
- Develop targeted retention campaigns for inactive and high-risk customer segments.
- Implement proactive monitoring systems using engagement and product utilization metrics as early warning indicators.
- Prioritize retention efforts for high-value customers to minimize revenue loss.

Conclusion

This project demonstrates that customer engagement and product utilization are significant drivers of customer retention in European banking. By leveraging data analytics and interactive dashboard reporting, financial institutions can identify at-risk customers earlier and implement targeted retention strategies. The findings support a more proactive, customer-centric approach to reducing churn and improving long-term business performance.

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Tools Used: Python, Pandas, NumPy, Plotly, Streamlit, Exploratory Data Analysis (EDA)